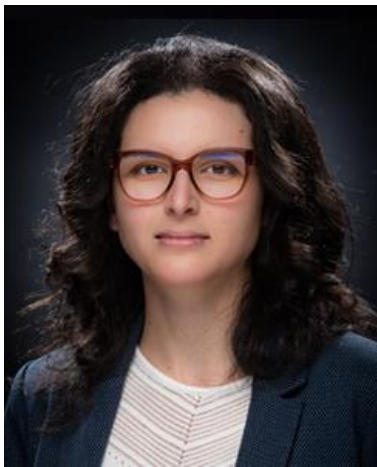




Dr. Eng. Lamice Denguir
CEO of Knowledge & Skills SAS, former
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Host: Dr. Jose C. M. Outeiro

Industrial Case Studies of Material Constitutive Modelling for Functional Performance and Lifespan Simulation

This presentation will delve into the application of material constitutive modeling in predictive simulation for sustainable products. It explains through two case studies, how sustainability issues in industry are prevented by predictive modeling, and the crucial role material constitutive models play in the accuracy of the predictions of functional performance and lifespan. The first case study, from an automotive manufacturer, will examine the development of thermo-mechanical constitutive models, enabling the simulation of fatigue resistance and the prevention of premature failure. We will see step by step how engineers identify laws ruling the thermo-mechanical behavior of materials from which exhaust manifolds in automotive heat engines are made, and use them to simulate fatigue resistance. The second case study will be from the nuclear industry. It shows how an enhanced predictive model of surface integrity induced by the manufacturing process, incorporating the accurate constitutive model for the work material, could enable the manufacturer to optimize his process parameters to enhance the corrosion resistance, and thus the safety of a critical component. These examples highlight the crucial role of material constitutive modeling in addressing sustainability challenges and ensuring the longevity and reliability of industrial components.

Bio: Dr. Lamice Denguir has a PhD in Mechanical Engineering, graduated from Arts et Metiers Institute of Technology, and is specialized in Manufacturing Processes. She led a diverse career in Europe spanning academia, industry, and entrepreneurship. As CEO of Knowledge and Skills Inc., she founded in 2020 the startup, managing, securing funding and launching innovative web platforms destined to engineers and scientists in engineering fields. Prior to this, she served as a scientific researcher at Karlsruhe Institute of Technology in Germany, Laboratoire Bourguignon des Matériaux et Procédés and Institut Carnot de Bourgogne in France, collaborating on international research projects and developing numerical models. Her consulting experience with Faurecia Clean Mobility involved material characterization, modeling, and simulation for automotive applications. Throughout her career, Dr. Denguir has led research projects, contributed to scientific publications, and earned recognition and awards like the valedictorian Medal of the Tunisian Association of Mechanics ATM (Tunisia 2012), CIRP Research Affiliate (2017), the Scientific Prize for best PhD thesis in manufacturing processes MUGV-Manuf21 (France 2018), and EdStart Innovators grant by AWS in European region for the startup Knowledge & Skills (France 2021).